

Ruhaan Bhargav

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EDUCATION

Purdue University

West Lafayette, IN

B.S. in Computer Science and B.S. in Mathematics, GPA: 3.74/4.0

Aug. 2024 – May 2028

- **Relevant Courses:** Data Structures & Algorithms, Systems Programming, Linear Algebra, Multivariate Calculus, Real Analysis, Differential Equations, Statistics

PROJECTS

Custom Ticketing Platform – Purdue Grand Prix

Dec. 2025 – Present

Python, SQLite, Flask, QR, Cryptography

- Architected a full-stack ticketing system with a normalized SQLite backend, QR code generation for contactless entry, and a secure authentication layer using cryptographic hashing and custom ID generation algorithms; replacing manual check-in processes for a 3,000-attendee event
- Designing a modular, extensible codebase to support API integration, database scaling, and Stripe payment integration; collaborating with a 2-person team using Git version control

High-Performance Order Book Matching Engine | *Python, Alpaca API, NumPy*

Nov. 2025 – Present

- Building a low-latency order book simulator processing 10,000+ orders/second with price-time priority matching; benchmarking heap, sorted list, and deque-based priority queue implementations for throughput optimization
- Implementing multivariate Hawkes processes for synthetic order flow generation; integrated Alpaca API for real-time SPY data ingestion to validate simulator accuracy against live market conditions

EXPERIENCE

Purdue University – CS Department

Aug. 2025, Mar. 2026 – Present

Student TA → Developer TA (rehired)

West Lafayette, IN

- Served as a TA for Purdue's intensive 2-week CS Bridge crash course (Aug. 2025), supporting students in foundational programming and problem-solving
- Rehired as Developer TA, authoring challenge problems, developing course material, and building automated test suites using JUnit to support the CS Bridge curriculum

UNICA China

Jun. 2025 – Aug. 2025

Software Engineer (Web)

Shanghai, China

- Developed and deployed a responsive, production-ready web interface using HTML, CSS, and JavaScript, automating client workflows and improving brand visibility (unicachina.com)
- Collaborated with cross-functional teams in Shanghai to deliver a mobile-compatible, desktop-first design; optimized image loading via lazy loading and compression, reducing load times by up to 5x

Purdue Orbital

Jun. 2024 – Dec. 2025

Avionics Engineer

West Lafayette, IN

- Designed and implemented embedded sensor packages (accelerometers, barometer, IMU) for rocket and high-altitude balloon telemetry from scratch using Rust and Embassy; published code adopted externally by other rocketry clubs including Stanford SSI
- Established a real-time inter-module data pipeline across GNC, Propulsion, and Systems teams using CAN bus protocol for structured telemetry communication

BoilerQuant

Jan. 2026 – Present

Quantitative Analyst

West Lafayette, IN

- Building data pipelines and statistical models to analyze time-series market data and identify pricing inefficiencies across equity and prediction markets and Co-authoring research about longshot bias in mid-size prediction markets

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Rust, JavaScript, C#, R, HTML/CSS, SQL

Libraries & Frameworks: PyTorch, pandas, NumPy, matplotlib, seaborn, scipy, Next.js, React.js, JUnit, Embassy

Tools & APIs: Git, SQLite, LaTeX, Linux, Alpaca API, yfinance, REST APIs

Spoken Languages: English (Primary), Mandarin (Conversational), Hindi (Native), Punjabi (Native)